The Power Spectral Density

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Key Concepts

1) The power spectral density characterizes a stationary random process in the frequency domain.

    a) The power spectral density is the discrete-time Fourier transform of the auto correlation sequence

    b) The area under the power spectral density between frequencies ω_a and ω_b is the average contribution to the power of the random process due to components with frequencies between ω_a and ω_b.

c) The total power or variance of the random process is the area under the power spectral density.

2) The power spectral density is non-negative.

3) The power spectral density is always real.

4) The power spectral density for a real-valued process is an even function of frequency.

5) Be careful about normalization - it is defined as a density. The units are power per radian or power per Hz.

a) Scaling the frequency axis from radians to Hz implies the amplitude must also scale.

b) Sometimes it may be normalized only on positive frequencies rather than positive and negative frequencies.

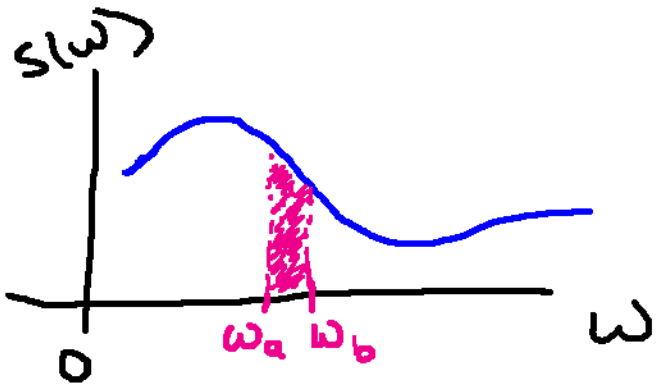
Video Screenshots

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Power spectral density characterizes a stationary random process in the frequency domain

$$S(\omega) \xleftrightarrow{\text{DTFT}} r[k]$$

$$S(\omega) = \sum_{k=-\infty}^{\infty} r[k] e^{-jk\omega} \qquad r[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S(\omega) e^{jk\omega} \, d\omega$$



$$P_{ab} = \frac{1}{2\pi} \int_{\omega_a}^{\omega_b} S(\omega) \, d\omega$$

avg contribution to total power (variance) due to components of the random process between ω_a and ω_b

avg ⇔ expected

Properties

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1) Units $S(\omega)$: power/radian $S(f)$: power / Hz

Note: $S(f) \neq S(\omega)\big|_{\omega = 2\pi f T}$ due to density

2) $r[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} S(\omega)\, d\omega = E\{x^2[n]\}$ (total power)

3) $S(\omega) \geq 0$ non-negative

4) $S(\omega)$ is real (follows from $r[k] = r^*[-k]$ and DTFT)

5) If $x[n]$ is real, $S(\omega)$ is even : $S(-\omega) = S(\omega)$

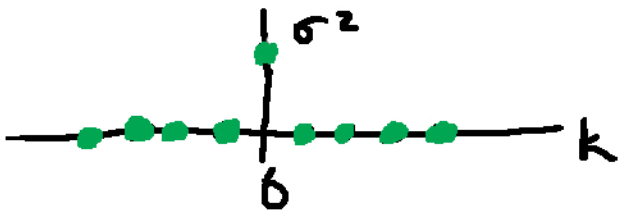
Caution: $S(\omega)$ may be normalized on $[0, \pi]$ rather than $[-\pi, \pi]$
$S(f)$ may be normalized to $[0, f_s/2]$ rather than $[\frac{-f_s}{2}, \frac{f_s}{2}]$

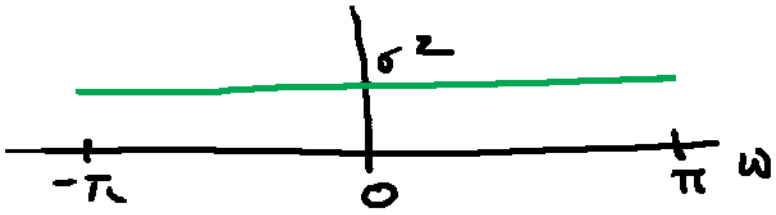
Examples

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1) White noise

$$r_w[k] = \begin{cases} \sigma^2 & k=0 \\ 0 & k\neq 0 \end{cases} \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad S_w(\omega) = \sigma^2$$





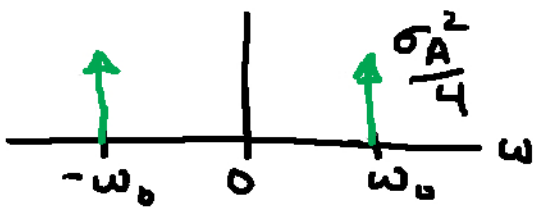
2) Random Sinusoid

$$s[n] = A\cos(\omega_0 n + \phi)$$

ϕ: uniform $[0, 2\pi]$

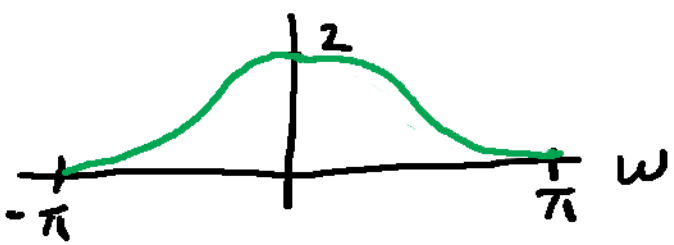
A: Gaussian, $E\{A\}=0$, $E\{A^2\}=\sigma_A^2$

$$r_s[k] = \frac{\sigma_A^2}{2}\cos(\omega_0 k) \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad S_s(\omega) = \frac{\sigma_A^2}{4}\delta(\omega+\omega_0) + \frac{\sigma_A^2}{4}\delta(\omega-\omega_0)$$



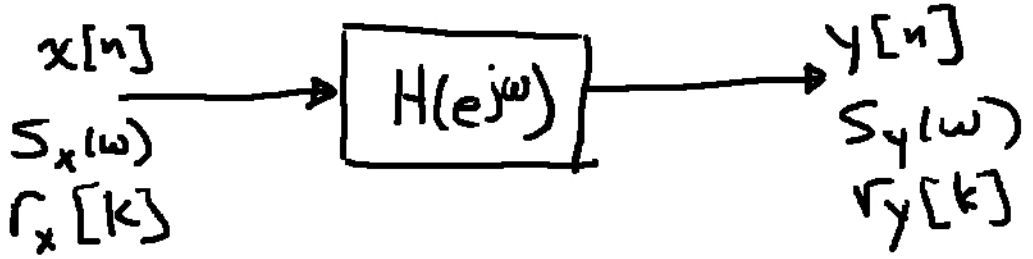
3) Colored Noise

$$r_c[k] = \begin{cases} 1 & k=0 \\ 1/2 & k=\pm 1 \\ 0 & \text{otherwise} \end{cases} \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad S_c(\omega) = 1 + \cos(\omega)$$



PSD and LTI Systems

6:01



$$y[n] = \sum_{m=-\infty}^{\infty} h[m]\, x[n-m]$$

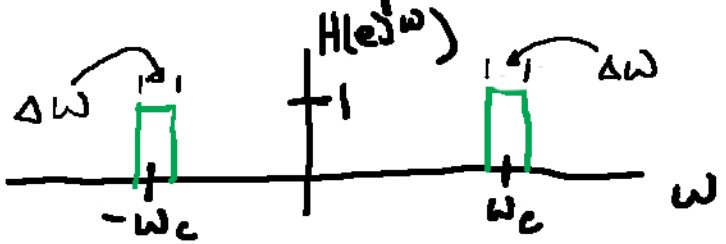
$$r_y[\ell] = E\{y[n]\, y[n-\ell]\} = \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} h[m]\, h[k]\, \underbrace{E\{x[n-m]\, x[n-\ell-k]\}}_{r_x[k+\ell-m]}$$

$$S_y(\omega) = |H(e^{j\omega})|^2\, S_x(\omega)$$

Power Spectrum Analyzer



ideal bandpass filter



$$P_y = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_y(\omega)\, d\omega \approx \frac{1}{2\pi} \left[S_x(-\omega_c)\Delta\omega + S_x(\omega_c)\Delta\omega \right]$$

$$= \frac{\Delta\omega}{\pi}\, S_x(\omega_c)$$

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